Md Riad Khan

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EDUCATION

Bangladesh Maritime University (BMU) Dhaka, Bangladesh
Faculty of Engineering and Technology Apr. 2024 – Apr. 2026
M.Sc. in Naval Architecture and Offshore Engineering CGPA: 3.91/4.00

- **Thesis**: Design Optimization and Performance Evaluation of an STHX for Marine Cooling Applications
  - Numerically evaluated the thermal performance and erosion characteristics of an STHX by simulating realistic seawater conditions.- Proposed an optimized STHX design and a cost-effective filtration strategy for marine cooling applications

Bangladesh University of Engineering and Technology (BUET) Dhaka, Bangladesh
Faculty of Mechanical Engineering (ME) May. 2012 – Feb. 2017
B.Sc. in Naval Architecture and Marine Engineering (NAME) CGPA: 3.77/4.00 (Class Position: 3/57)

- **Thesis**: Energy Study: Contra-rotating Propeller System
  - Developed a computer program to calculate the self and mutually induced velocities of a Contra-rotating Propeller (CRP) system and predicted its efficiency using both Blade Element Theory and Circulation Theory. Compared the predicted CRP efficiency with that of conventional single-screw propeller systems

AREAS OF INTEREST

- Robotics & AUV
- AI & ML
- Heat Transfer
- Computational Fluid Dynamics
- Mechanics of Structure
- Engines and Combustion

TECHNICAL SKILLS

- **Programming Languages:** Python, MATLAB, Fortran, C++
- **Simulation:** ANSYS Fluent, ANSYS Mechanical, Maxsurf, SHIPFLOW
- **Design Software:** SolidWorks, Rhino 3D, AutoCAD
- **Document Editing Tools:** Overleaf (LaTex), MS Office

RESEARCH & TEACHING EXPERIENCE

Bangladesh Maritime University (BMU) Dhaka, Bangladesh
Lecturer, Department of Naval Architecture and Offshore Engineering (NAOE) Sep. 2023 – Present

- **Offered Courses**: Heat Transfer (scheduled, Fall 2026), Mechanics of Structure, Engines and Fuels, Hydrostatics and Stability, Introduction to NAOE, Mechanical Engineering Drawing and CAD Lab, Ship Design Lab 1, Numerical Methods in Engineering Lab (scheduled, Fall 2026)
- **Undergraduate Thesis Supervision**: Supervised Undergraduate theses on CFD, Robotics, Autonomous Vehicles, Image Processing, Combustion Processes, etc.
- **Reviewer**: Journal of Engineering and Technology (JET), Bangladesh Maritime Journal (BMJ)
- **Club Coordinator**: BMU Cultural Club

SELECTED PUBLICATIONS

Journal Articles

- **Khan, M. R.**, Kundu, R., Shah, R. I., & Hasan, R. (2025). Predicting a vessel's trajectory: A simple machine learning method. *Amesia*, 6(2), 122129. https://doi.org/10.54559/amesia.1764517 (Link)
- Kundu, R., **Khan, M. R.**, Rahman, S., & Saha, G. K. (2024). Marine heat waves in a low-lying country: Occurrence, computation and impact. *Journal of Naval Architecture and Marine Engineering*, 21(2), 117124. https://doi.org/10.3329/jname.v21i2.75296 (Link)
- **Khan, M. R.**, Sharmin, N., & Hasan, S. M. R. (2025). Mobile wave crawler for energy generation: Preliminary feasibility and opportunities for Bangladesh. *Journal of Bangladesh Academy of Sciences*, 49(2), 215225. https://doi.org/10.3329/jbas.v49i2.81329
- Talokder, M. A. R., Rahman, M. M., Haque, M. M.,**Khan, M. R.** (2022). Compatibility of the pusher-barge system in inland and coastal waterways transportation system of Bangladesh. *Journal of Maritime Research*, 19(2), 3943

Journal Articles Under Review/ Preparation

- **Khan, M. R.**, Hasan Roni, M. R., Karim, M. M. & Miljkovic, N. Numerical investigation of the impact of suspended solids in seawater on a shell and tube heat exchanger in marine cooling systems and development of a cost-effective filtration system
- **Khan, M. R.**, Hasan Roni, M. R., & Karim, M. M. Numerical Assessment of Advanced Baffle Arrangements for Improved Heat Transfer and Flow Distribution in a Shell-and-Tube Heat Exchanger
- Khan, M. M., Abir, M. J. R., **Khan, M. R.**, & Islam, S. Deep learning framework for morphological identification of Hilsa in high-turbidity riverine ecosystems
- Hasan, S. M. R., Karim, M. M., Hossain, K. A.,**Khan, M. R.**, Islam, M. S., & Raihan, A. CFD-based validation of fuel-saving ship hull design methods for inland ships. Accepted in *Scientific Reports* (PDF)

Conference Proceedings

- **Khan, M. R.**, Kundu, R., Rahaman, W., Haque, M. M., & Ullah, M. R. (2018). Efficiency study: Contra-rotating propeller system. In *AIP Conference Proceedings* (Vol. 1980, No. 1, Article 060001). International Conference on Mechanical Engineering (ICME 2017), Dhaka, Bangladesh. https://doi.org/10.1063/1.5044369 (Link)
- Kundu, R., **Khan, M. R.**, Rahman, S., & Saha, G. K. (2018). Design and structural analysis of a pontoon vessel. In *AIP Conference Proceedings* (Vol. 1980, No. 1, Article 030006). International Conference on Mechanical Engineering (ICME 2017), Dhaka, Bangladesh. https://doi.org/10.1063/1.5044285 (Link)
- Rahman, W., Sworan, M. R. I., **Khan, M. R.**, & Baree, M. S. (2018). Structural analysis of a container vessel hatch cover by finite element method. In *AIP Conference Proceedings* (Vol. 1980, No. 1, Article 030005). International Conference on Mechanical Engineering (ICME 2017), Dhaka, Bangladesh. https://doi.org/10.1063/1.5044284 (Link)
- Khaled, M. E., Kawamura, Y., Abdullah, M. S. B., Banik, A., Faruk, A. K., & **Khan, M. R.** (2018). Assessment of collision & grounding risk at Chittagong Port, Bangladesh. In *Proceedings of the 11th International Conference on Marine Technology*, Kuala Lumpur, Malaysia. (PDF)
- **Khan, M. R.**, Islam, M. A., & Abdullah, A. (2020). Time and cost effective ship design process using single parent design approach with considerable dimension difference. In *Proceedings of the 12th International Conference on Marine Technology*, Indonesia.
- Abdullah, A., Moinuddin, A., **Khan, M. R.**, & Islam, M. A. (2020). Advancement of ship recycling industry of Bangladesh in comparison to the rest of the world with obstacles, root causes and propositions. In *Proceedings of the 12th International Conference on Marine Technology*, Indonesia.

PROJECTS/ RESEARCH GRANTS

- Project Director: **Assessing Domestic Industrial Capabilities to Support Shipbuilding Industries in Bangladesh**, (Jan. 2026 - Present), Funded by: UGC
- Consultant Team Member: **Feasibility Study for the Procurement and Installation of 4 Winch Barges with Ancillary Equipment to Increase the Salvage Capacity of the Salvage Unit**, (Aug. 2025 - Jan. 2026), Project awarded to FET, BMU by Bangladesh Inland Water Transport Authority (BIWTA)
- Consultant Team Member: **Feasibility Study for Procurement of Dredge Material Carrying Dumb Barges**, (Jun. 2024 - Feb. 2025), Project awarded to FET, BMU by BIWTA
- Associate Investigator: **Sustainable Inland Shipping: Validating Fuel-Saving Ship Hull Design Methods for Inland Ships of Bangladesh**, (Jun. 2024 - Jun. 2025), Funded by: Ministry of Science and Technology, Bangladesh

PROFESSIONAL EXPERIENCE

Mongla Port Authority Mongla, Bagerhat, Bangladesh
Assistant Engineer Jun. 2022 – Sept. 2023

- **Key Responsibilities**: Project Supervision, Troubleshooting, Design, Drafting, Estimation, Maintenance

Khulna Shipyard Ltd (KSY), Bangladesh Navy Khulna, Bangladesh
Assistant Engineer Jul. 2017 – Jun. 2022

- **Key Responsibilities**: Project Management and Leadership; Design, Drafting, and Planning; Calculations and Cost Estimation; Tender Preparation and Procurement; Documentation and Presentation; Quality Assurance, Control, and Supervision; Commissioning, Testing, and Trials; Technical Review, Troubleshooting, and Problem Solving
- **Major Completed Projects**: Three Inshore Patrol Vessels (IPV) for Bangladesh Coast Guard; Two Survey Vessels for Bangladesh Navy

HONORS AND AWARDS

- **University Merit Scholarship:** Awarded for outstanding academic performance in the University Examinations (2012, 2013, 2015, and 2017), BUET
- **Dean's List:** Awarded for obtaining a CGPA above 3.75 (2012, 2013, 2015, 2016), Faculty of ME, BUET
- **Higher Secondary Certificate (HSC) Talent Pool Scholarship:** Awarded for outstanding performance in the HSC Examination under the Jessore Education Board, Bangladesh (2011 - 2015)
- **Appreciation Letter:** Received multiple Appreciation Letters for successful completion of projects at KSY

TRAININGS

- Training on **SHIPFLOW** by Flowtech Int. (Oct. 2025), BMU
- Smart Port Management: Empowering Economic Prosperity with **AI-Based Port** Research Bootcamp by Dr Burak Cankaya, Associate Professor, Embry-Riddle Aeronautical University, USA (7 - 11 Sep. 2025), BMU
- Training on Competency-Based Education and Training (**CBET**), (10 - 13 Mar. 2024), BMU
- Training on **Engine, Z-Peller and Remote Control System**, (18 Nov. 2022), IHI Power Systems Co. Ltd., Japan
- Course on **Japan-Asia Youth Exchange Program in Science**, (11 - 20 Dec. 2016), Osaka University, Japan

PROFESSIONAL AFFILIATIONS

- Associate Member of **Institution of Engineers, Bangladesh (IEB)**, ID: A21349
- Life Member of **Association of Naval Architects and Marine Engineers, Bangladesh (ANAMEB)**, ID: LM106220

EXTRA-CURRICULAR ACTIVITIES

- Served as a **Drummer** in local musical bands, including Cyanic Dearth, Traum, and Surreal
- Performed as an **Acoustic Guitarist** at cultural programs and community events
- Actively participated in organizing cultural, social, and other extracurricular events
- Provided private tutoring in advanced mathematics, physics, and chemistry to high school and college students.

REFERENCES

Dr. Md. Sadiqul Baree
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Department of NAOE, BMU
Ex-Professor & Head
Department of NAME, BUET
Email: sadiqul.naoe@bmu.edu.bd

Dr. Md. Rakibul Hasan Roni
Assistant Professor
Department of Mechanical Engineering
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and Technology (BUET)
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